Riemann Sums and Definite Integrals

Section 1: Left and Right Riemann Sums

1 Approximate $\int_0^4 x^2\,dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14

b. 30

c. 21

d. 16

2 Approximate $\int_0^4 x^2\,dx$ using a Right Riemann sum with $n = 4$ equal subintervals.

a. 30

b. 14

c. 22

d. 28

3 For a strictly increasing function $f(x)$ on $[a, b]$, which inequality correctly relates the exact integral I to the Left (L_n) and Right (R_n) Riemann sums?

a. $L_n < I < R_n$

b. $R_n < I < L_n$

c. $I < L_n < R_n$

d. $L_n = I = R_n$

Section 2: Midpoint and Trapezoidal Approximations

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4 Approximate $\int_1^5 \frac{1}{x}\,dx$ using the Midpoint Rule with $n = 2$ equal subintervals.

a. $\frac{1}{2} + \frac{1}{4} = 0.75 \approx 1.5$ is wrong; width=2: $2(1/2 + 1/4) = 1.5$

b. 1.5

c. 1.2

d. 1.33

5 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^2 x^3\,dx$.

a. 4.25

b. 4.00

c. 3.75

d. 4.50

6 If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal approximation T_n compare to the exact integral I?

a. $T_n > I$ (overestimate)

b. $T_n < I$ (underestimate)

c. $T_n = I$

d. Cannot be determined without $f'(x)$

Section 3: Definite Integral as the Limit of a Riemann Sum

7 Express the limit $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^4 \frac{3}{n}$ as a definite integral.

a. $\int_2^5 x^4\,dx$

b. $\int_0^3 (2 + x)^4\,dx$

c. $\int_2^5 (2 + 3x)^4\,dx$

d. $\int_0^1 (2 + 3x)^4\,dx$

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8 Evaluate $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \frac{i^2}{n^2}$.

a. $1/3$

b. $1/2$

c. $1/4$

d. 1

9 Convert $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n+i}$ into a definite integral on $[0,1]$ and evaluate it.

a. $\ln 2$

b. 1

c. $\ln 3$

d. $1/2$

Section 4: Geometric Evaluation of Definite Integrals

10 Evaluate $\int_{-3}^{3} \sqrt{9-x^2}\,dx$ using geometry.

a. $\frac{9\pi}{2}$

b. 9π

c. $\frac{9\pi}{4}$

d. 18

11 Evaluate $\int_{0}^{4} (2x-3)\,dx$ by computing net signed area.

a. 4

b. 8

c. 0

d. -2

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12 If $\int_1^5 f(x)\,dx = 7$ and $\int_3^5 f(x)\,dx = 4$, find $\int_1^3 f(x)\,dx$.

a. 3

b. 11

c. -3

d. 7/4

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